

Project Manual

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Description

Carbon monoxide (CO) is a poisonous gas that is both odorless and colorless, making it undetectable to human senses. Being exposed to too much of it can cause serious illness, or even death. Because of this, we are creating a system that will use sensors to detect CO levels in an area and create a high or low output, depending on the amount detected. In milestone 1, we created a system to light up an LED if unsafe levels of CO were detected. In milestone 2, we built on top of this, adding a ventilation system. When unsafe levels of CO are detected, the ventilation system turns on. Because higher levels of CO will take longer to bring down, we implemented a way to increase fan speed the longer the ventilation system is active. This means that higher levels of CO will end up resulting in higher fan speeds, and lower levels won't require this. This is our way of creating energy efficiency in our design.

The block diagram for milestones 1 and 2 is shown below in figure 1. Here we are using an op amp as a comparator to do our basic decision making: if unsafe levels of CO are detected then turn on the LED and if not, the LED stays off. We have two inputs going into the op amp. The first is the output from the sensor after passing through an RC Low Pass Filter. The second is our reference voltage that we got by using a potentiometer. This reference voltage will equal the voltage of the output when the sensor detects 50 parts per million (ppm) which was a value we sourced from OSHA [1]. The output from the comparator will then be connected to an LED which will turn on when the sensor voltage is greater than the reference voltage then turn off when the sensor voltage is less than the reference voltage. In milestone 2, we use this same output from the comparator. This high/low voltage is fed to an op amp with capacitive feedback. If there is a high voltage input from the comparator in milestone 1, the integrator will begin to output a ramping voltage. The output of the integrator is connected to a 555-timer used as an astable multivibrator. The astable multivibrator creates a voltage pulse and as the voltage ramp increases, the duty cycle also increases. The duty cycle is the ratio of the time the astable multivibrator is operating to the total time of one cycle. The output of our astable multivibrator is fed to a DC brushless fan which is acting as our ventilation system. The higher the duty cycle, the faster the DC motor in the fan will spin.

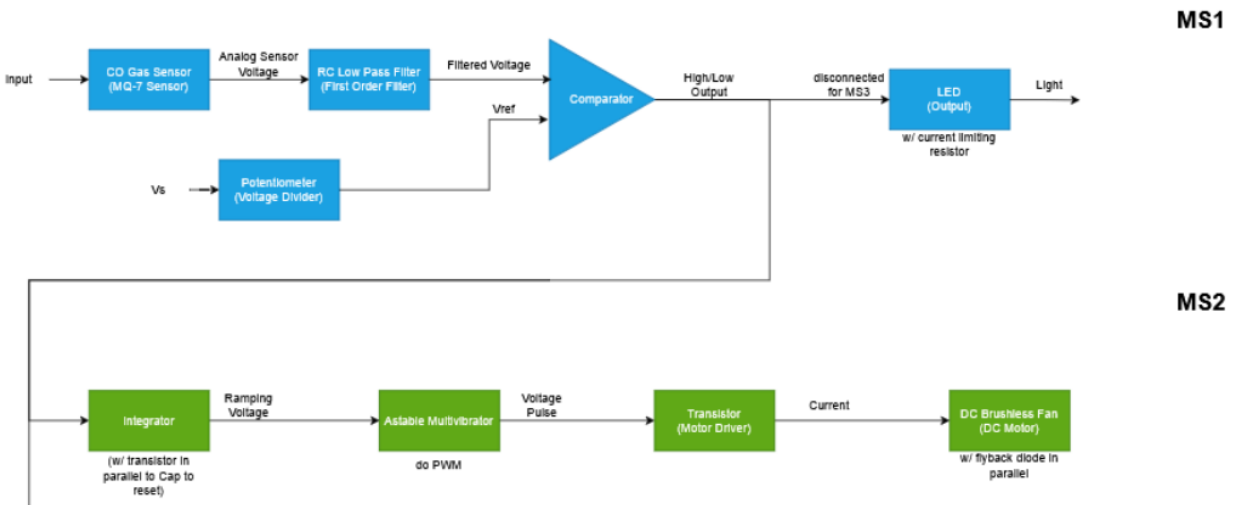


Figure 1: Complete Block Diagram for Milestone 1 and 2

The complete schematic for milestones 1 and 2 is shown below in figure 2. We have two 5-volt voltage sources to power our op amp and provide our input voltages for the op amp. The +5V source is the voltage source to get our reference and sensor voltage. Two 10k potentiometers were used in milestone 1 to set the reference voltage which was left untouched and to simulate the sensor voltage which was done by turning the knob. For example, we can simulate high levels of CO by turning the knob one way to make R_{s1} small and turn it the other way making R_{s1} large to simulate low levels of. This sensor voltage is then put through the low pass RC filter to filter any noise from the voltage output of the sensor. This is based on the property of a capacitor where it is unable to change voltage instantaneously in cases where the signal frequency is high and only when signal frequency is low can the change be noticeable. To compare the voltages, the reference voltage is connected to the negative input of the op amp, and the filtered sensor voltage is connected to the positive input. Since an ideal op amp has infinite gain, if the sensor voltage is greater than the reference voltage, the output voltage will be saturated at V_{cc+} and if the sensor voltage is less than the reference voltage, the output voltage will be saturated to V_{cc-} . When we have a positive output, the LED will light up and if the output is negative, the LED will be off. For our integrator design in milestone 2, we use this output (V_{comp}) as the input to our system. As you can see below, this input is then brought to two different op amps. The top op amp is used as a comparator and is used to discharge the capacitor. When our milestone 1 system detects that the CO levels are safe, the output of the comparator will be equal to V_{cc-} (-5V). This low voltage is brought to the comparator and is compared to the reference voltage. Since in this scenario, the voltage reference is greater, and the reference voltage is connected to the positive input, the comparator outputs a high voltage. This output is connected to a NMOS transistor, and when high, connects the output of the integrator to negative input. In an ideal op amp, the voltage at the negative input equals the voltage at the positive input, so in this case, the negative input is basically ground. In conclusion, the capacitor is discharged when the NMOS transistor is active. In addition to this, V_{comp} is also brought to the lower op amp. This op amp is a non-inverting amplifier that acts as an inverter. So, when V_{comp} is a

high voltage, indicating unsafe CO levels, it will be a negative value with the same magnitude after this inverter. This output is connected to the negative input of our op amp with capacitive feedback. Using capacitive feedback, we can take advantage of the relationship between the current and voltage of a capacitor ($i_c = C \frac{dV_c}{dt}$) to create a component that integrates the input. This integration is what allows us to get our ramping voltage, which is then brought to the astable multivibrator. This is a 555 circuit that continuously charges and discharges a capacitor through resistors, producing repeated square waves whose duty cycle increases at the ramping voltage increases.

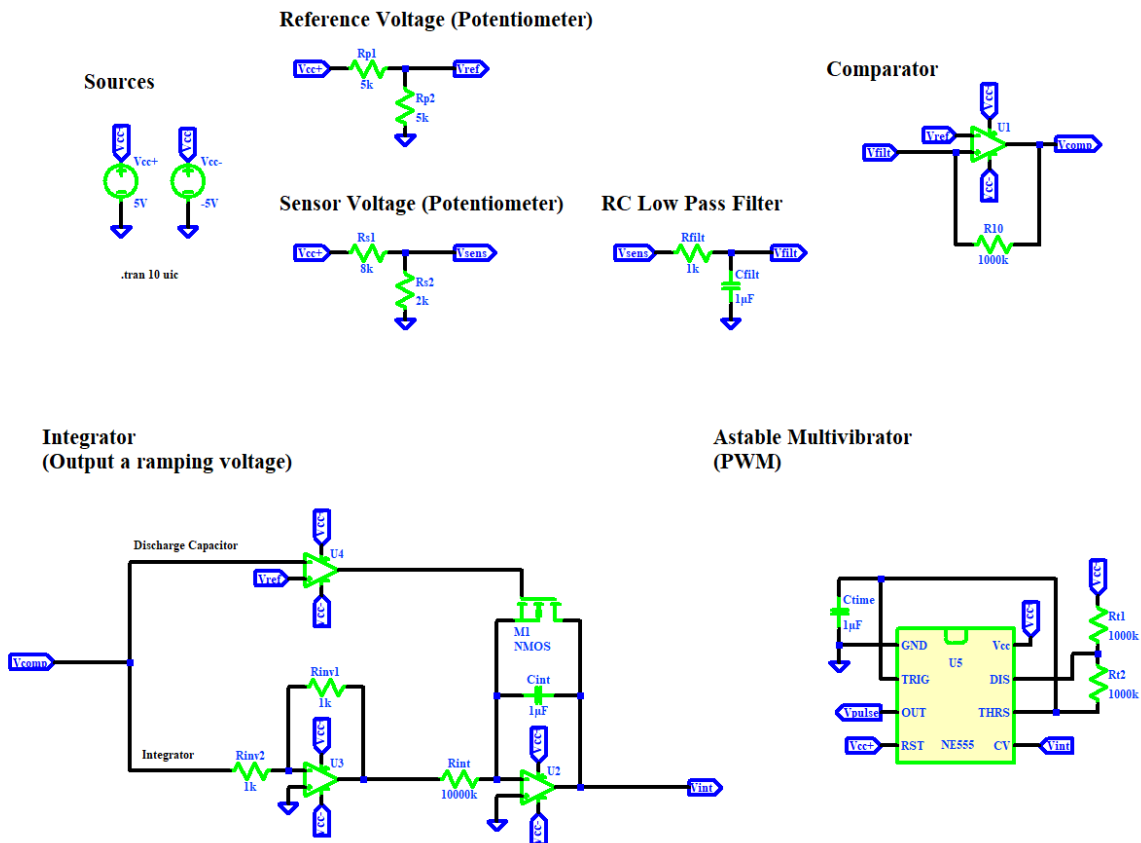


Figure 2: Complete Schematic for Milestone 1 and 2

A pre-existing application of a CO sensor is the household CO sensor by first alert. The sensor's output is typically a low-level analog voltage. Before getting to the decision-making stage, the circuit includes an op amp to amplify the signal to a readable voltage level and a low-pass RC filter. After this, it used a microcontroller to decide: ~50 ppm an LED indicator appears, ~150 ppm causes an alarm, ~400 ppm causes an immediate alarm indicating critical danger [3]. Since we are tasked to design our labs as entirely analog systems, we decided to use an op amp to do the decision making.

Operation and Design

Input, Potentiometer for Sensor Voltage

Potentiometer for Sensor Voltage

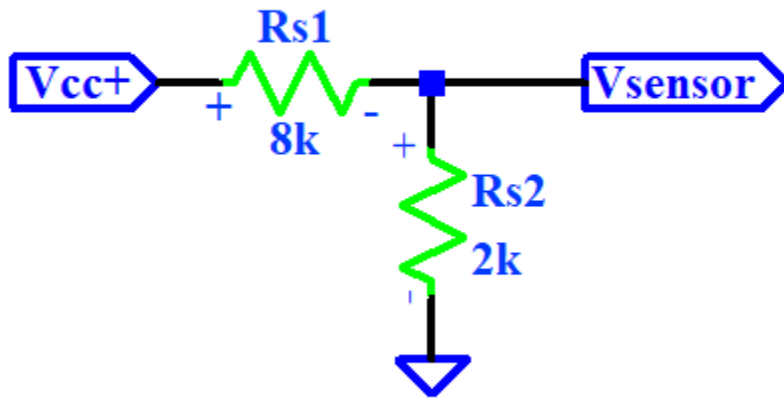


Figure 3: Input Block Schematic

Using the voltage divider equation:

$$V_{sensor} = V_{cc+} \left(\frac{R_2}{R_1 + R_2} \right) \quad (1)$$

Applying this formula to our specific input block gives us the equations relating to the input and output:

$$V_{sensor} = 5 \left(\frac{2k}{10k} \right) \quad (2)$$

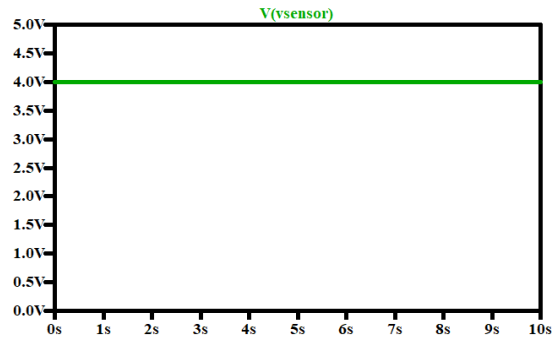
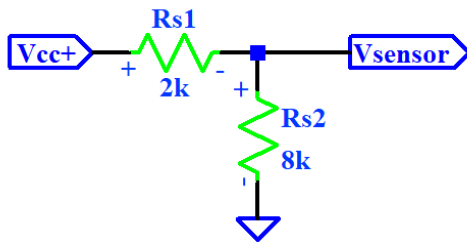
Since we vary this potentiometer to simulate different outputs from the CO sensor, here is a more general equation:

$$V_{sensor} = 5 \left(\frac{R_{s2}}{R_{s2} + R_{s1}} \right) \quad (3)$$

For our sensor voltage, since we are using it for a comparator, we simply simulate for results where the sensor voltage is greater than and when it's less than the reference voltage of 2.5V.

And so, to get a sensor voltage greater than 2.5V, we turn the knob to the right to change the resistance values of the voltage divider:

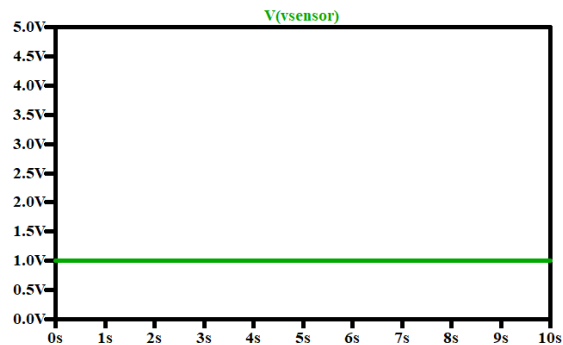
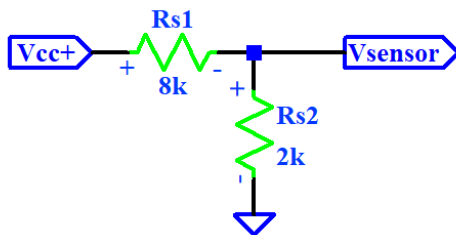
Potentiometer for Sensor Voltage



R_{s1} and R_{s2} have a relationship where $R_{s1} < R_{s2}$ now, the sensor voltage is greater than the reference voltage and in this case is set to 4V.

Now turning the knob to the left:

Potentiometer for Sensor Voltage



R_{s1} and R_{s2} now reverse their relationship to $R_{s1} > R_{s2}$ which results in the sensor voltage becoming less than 2.5V and in this case 1V.

MS1 Building Block, Comparator

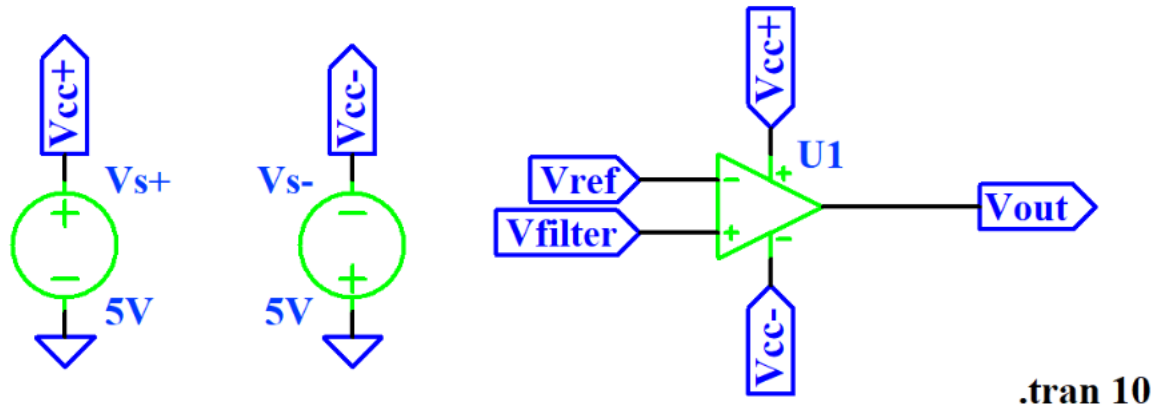


Figure 5: Op amp comparator schematic

We have the reference voltage connected to the negative input of the op amp and the filtered voltage from the sensor connected to the positive output of the op amp.

Because of this, the behavior of the op amp as a comparator is:

$$V_{ref} > V_{filter} \rightarrow V_{out} = V_{cc-} = -5V \quad (4)$$

$$V_{filter} > V_{ref} \rightarrow V_{out} = V_{cc+} = +5V \quad (5)$$

We chose this specific op amp design to allow our circuit to do very basic decision making. This design lets us set a reference voltage and turn on an LED if the other input is greater than this reference value. The specific op amp we chose for this design is the TL072IP from TI. There are two reasons we chose this: (1) it came in the kit that we bought and (2) it is an op amp designed to be used as a comparator.

The plot of input vs output for our op amp as a comparator is shown below in figures 6 and 7. It shows that when the filtered voltage is greater than the reference voltage, the output voltage is 5V and when the filtered voltage is less than the reference voltage, the output is -5V.

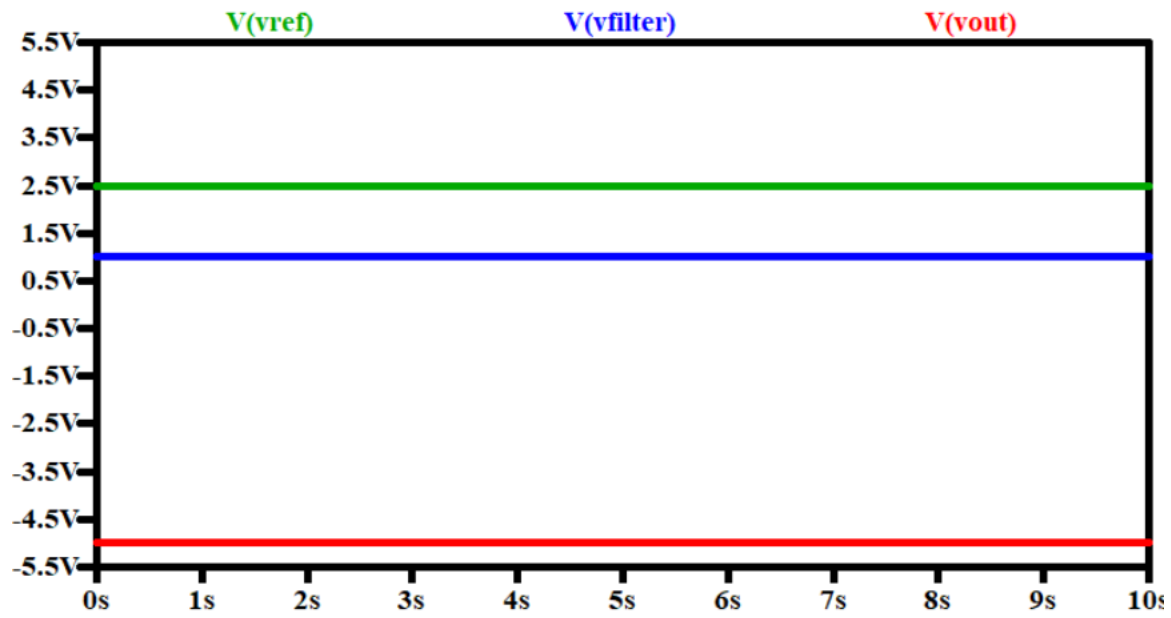


Figure 6: Comparator input vs output when $V_{ref} > V_{filter}$

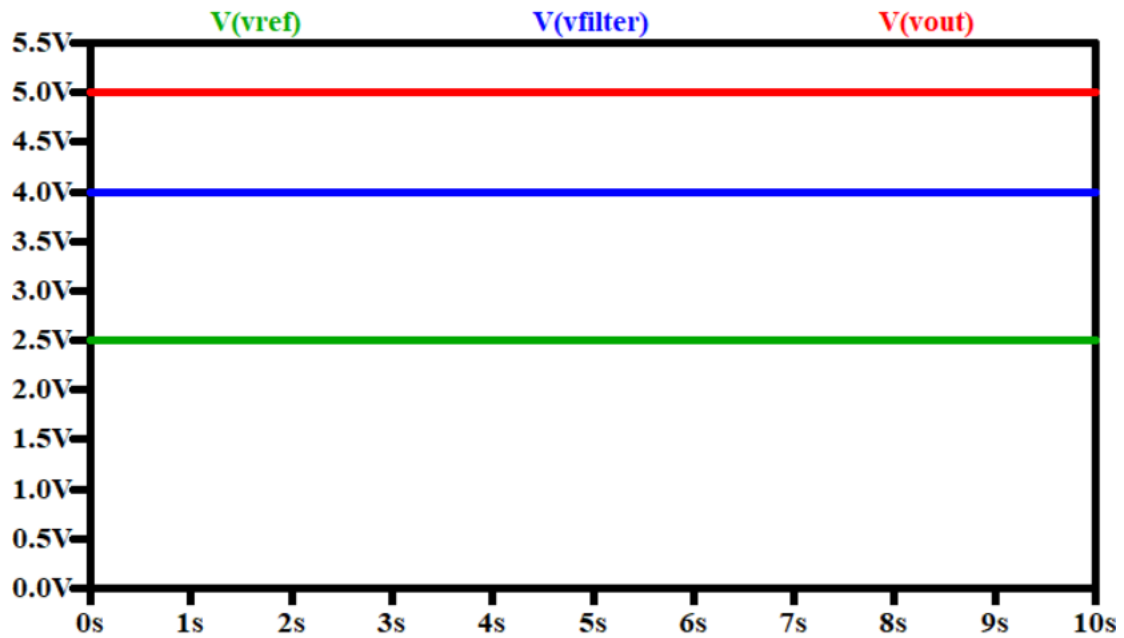


Figure 7: Comparator input vs output when $V_{filter} > V_{ref}$

MS1 Output, LED

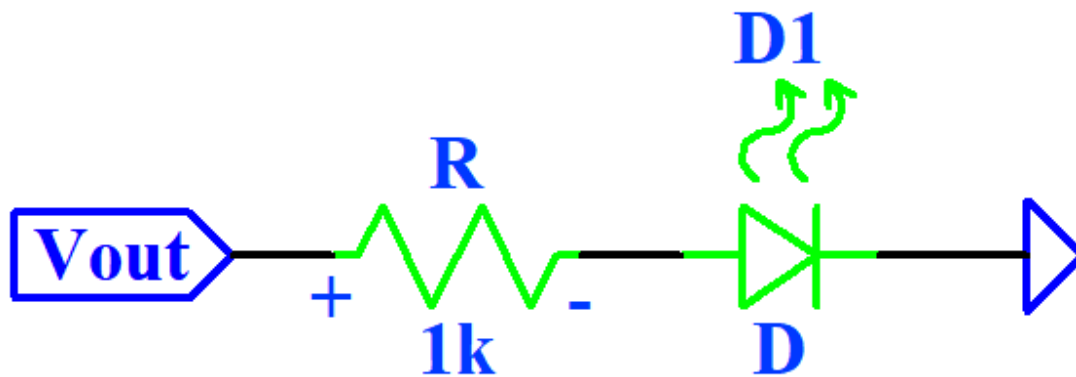


Figure 8: MS 1 Output block schematic

We are using a general 5mm (T-1 ¾) red LED that came in our parts kit. Based on the data sheet, this particular LED has a forward voltage range from 1.7-2.4V [2]. It also has a peak forward current of 100mA [2].

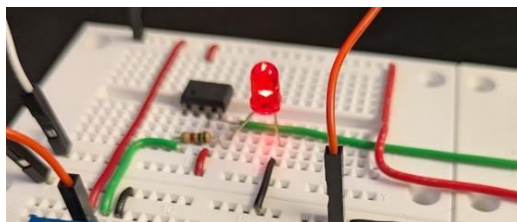
In figure 7, V_{out} is the output voltage from the op amp. So, it will only be 5V or -5V. Below are the two statements for the output:

$$V_{out} = 5V \rightarrow LED \text{ turns on} \quad (6)$$

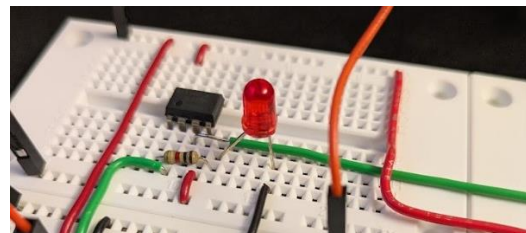
$$V_{out} = -5V \rightarrow LED \text{ turns off} \quad (7)$$

To demonstrate this functionality, we took photos of the LED for the cases where V_{out} is +5V and when V_{out} is -5V.

When our V_{out} is +5V, the LED is on:



And when V_{out} is -5V, the LED is off:



MS2 Building Block, Integrator

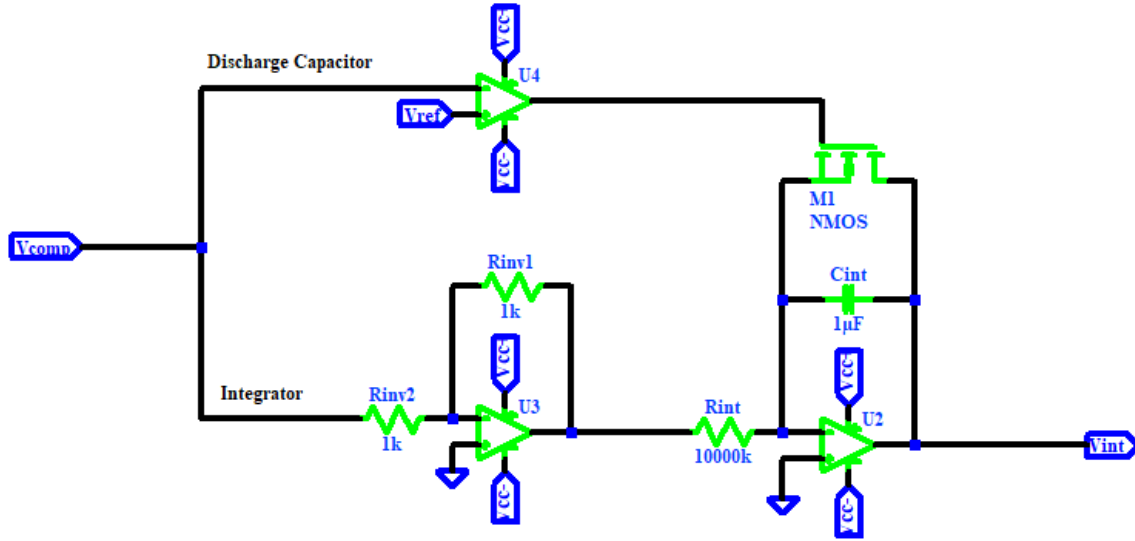


Figure 9: Integrator Schematic

Op amp to discharge the capacitor:

$$V_{ref} > V_{comp} \rightarrow V_{out} = V_{cc+} = 5V \quad (8)$$

$$V_{comp} > V_{ref} \rightarrow V_{out} = V_{cc-} = -5V \quad (9)$$

Since we use a non-amplifying inverter cascaded with an inverting integrator:

$$V_{int} = (-1) \left(-\frac{1}{R_{int} C_{int}} \right) \int V_{comp} dt \quad (10)$$

To make the integrator ramp in the time of 10s from 0V to 5V, we needed to choose specific values for C_{int} and R_{int} . Using the time constant equation, we found that if we use $C_{int} = 1 * 10^{-6}$, then $R_{int} = 1M\Omega$.

To show the operation of the integrator, the simulation shows how the integrator is triggered by the V_{comp} output from MS1 so when V_{comp} is HIGH, the integrator ramps from 0V to 10V in 10s.

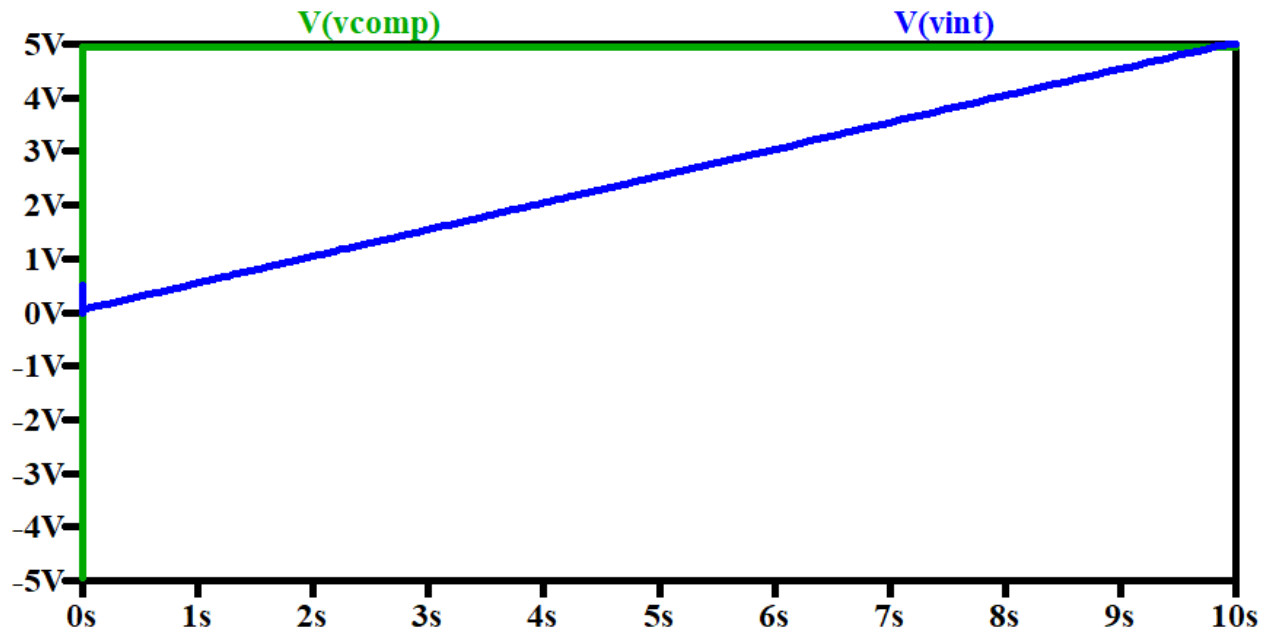


Figure 10: Integrator Output when V_{comp} is High

Then when V_{comp} is LOW, the integrator stays at 0V.

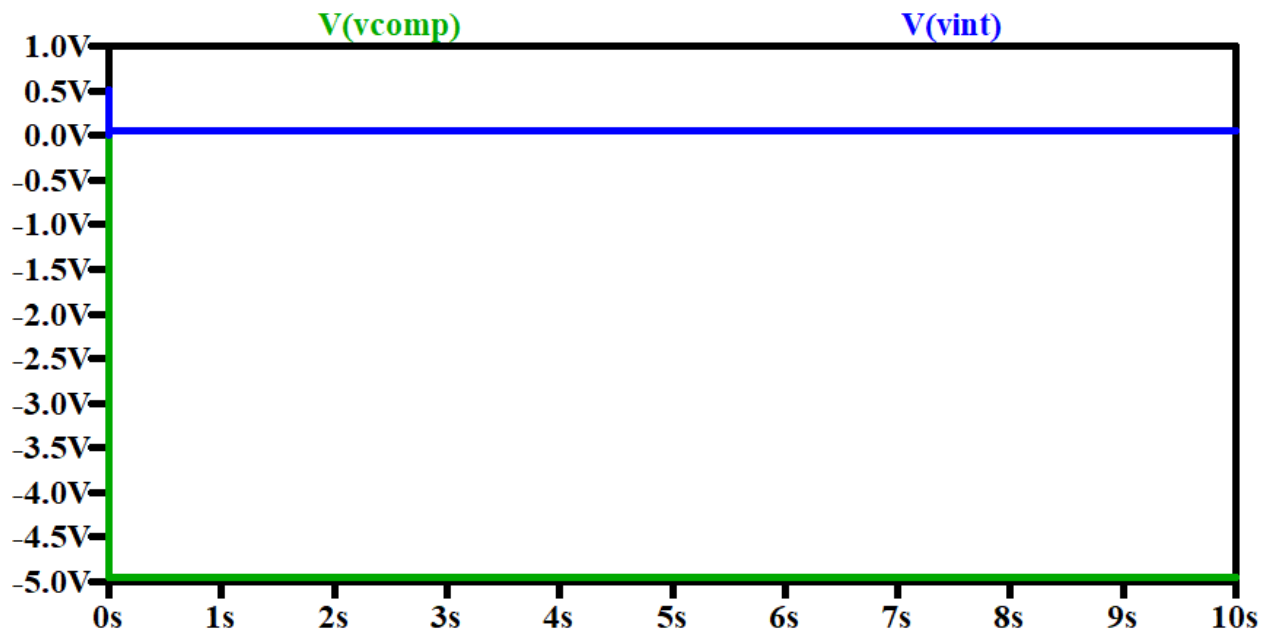


Figure 11: Integrator Output when V_{comp} is Low

Also, since we included a system to discharge the capacitor, we can display this by making V_{comp} cycle through HIGH and LOW in the simulation:

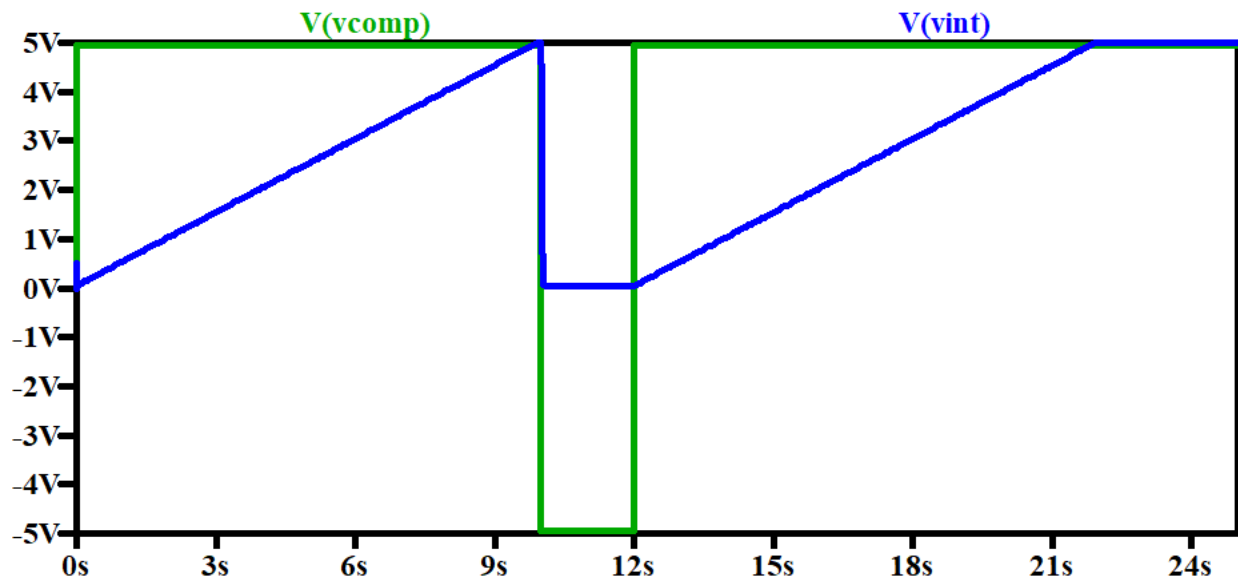


Figure 12: Integrator Output to Demonstrate Capacitor Discharge

So, when V_{comp} is HIGH, the integrator is ramping and when V_{comp} becomes LOW, the integrator becomes 0V and when V_{comp} is HIGH again, it starts ramping from 0V to 5V as desired.

MS2 Building Block, Astable Multivibrator

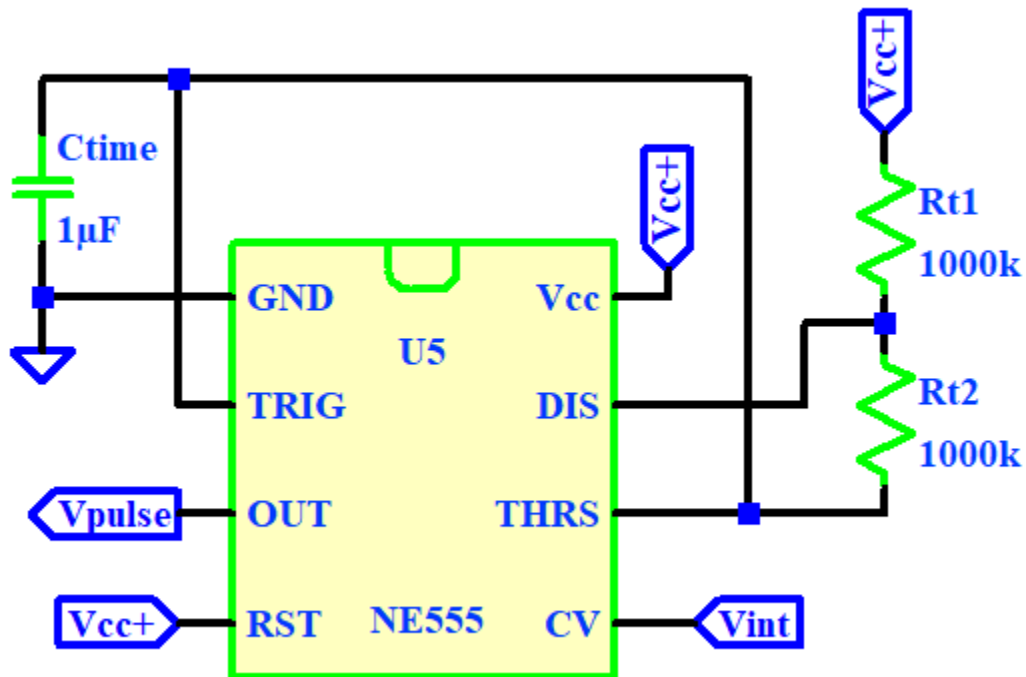


Figure 13: Astable Multivibrator Schematic

For the astable multivibrator, with an input at CV the time low remains the same but time high changes to have an increasing duty cycle:

$$t_{HIGH} = 2 \ln\left(\frac{5 - 0.5V_{int}}{5 - V_{int}}\right) \quad (11)$$

$$t_{LOW} = 0.693(R_{t2})C_{time} \quad (12)$$

We picked values of $C_{time} = 1 * 10^{-6}F$ and $R_{t1} = R_{t2} = 1M\Omega$ to have a period that is comparable to the 10s that the integrator takes to ramp from 0 to 5V

The action of the astable multivibrator relies on the ramping voltage from the integrator so when the integrator is ramping (V_{comp} is HIGH) then it outputs a pulse voltage with increasing duty cycle:

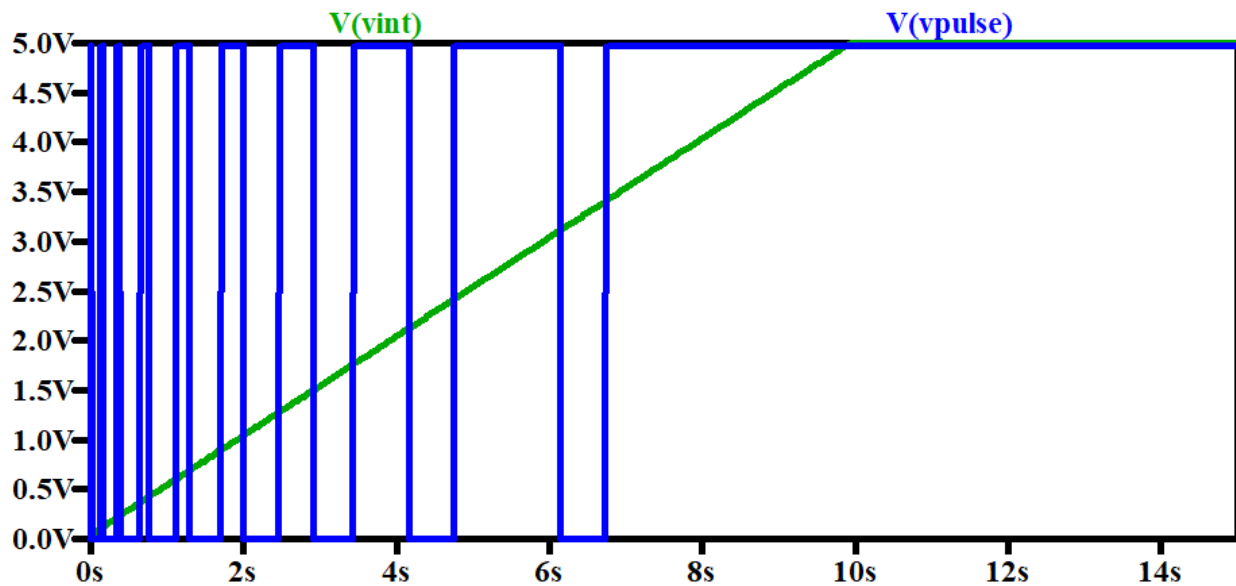


Figure 14: Astable Multivibrator Output When Receiving a Ramping Voltage

Then when the integrator is sitting at 0V (V_{comp} is LOW), the pulse output of the astable multivibrator is 0V.

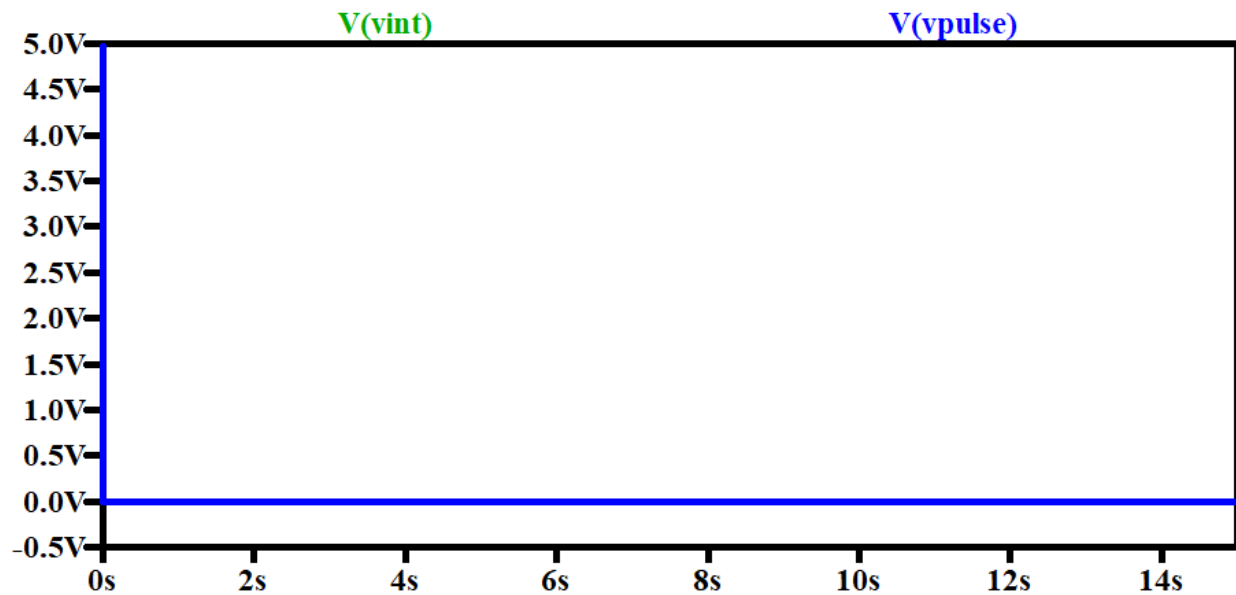


Figure 15: Astable Multivibrator Output When Receiving 0V

MS2 Output, DC Motor

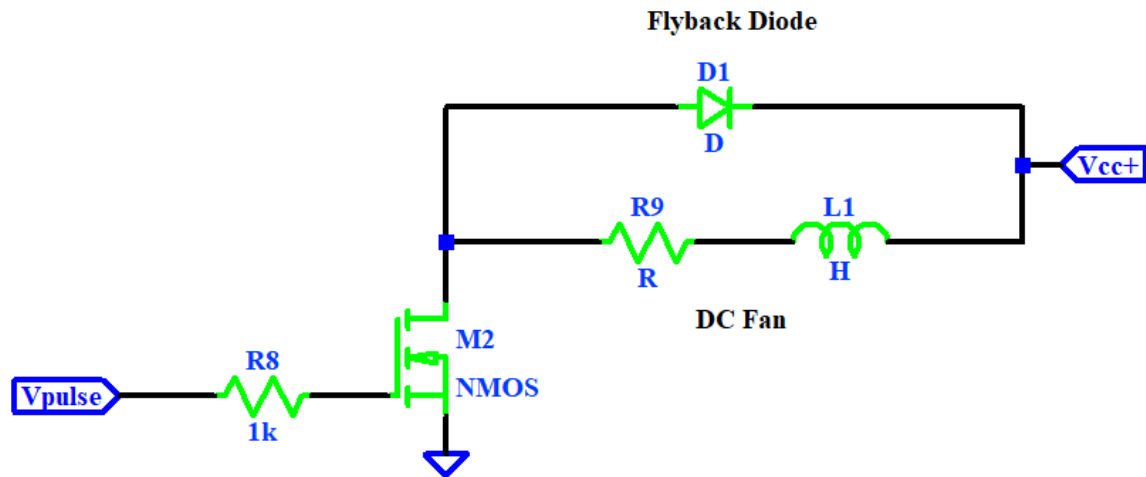


Figure 16: MS2 Output Block Schematic

The DC fan directly reflects the input of V_{pulse} so when V_{pulse} is triggered and outputs a pulse signal, the DC fan is on, otherwise when V_{pulse} is staying at 0V, the fan is off.

Since a DC motor is an inductor, we needed to use a flyback diode in parallel to the motor to protect the transistor from any backwards current that would be generated from the motor after turning off the pulse signal. Also, since the M2K board doesn't have enough current from its voltage sources, we used a +5V source from a microcontroller for the V_{cc}^+ connection to the fan which had over 0.2A.

Integration and Optimization

The first part of our milestone one is the input block. Since we didn't get the CO sensor in time, the sensor voltage is produced using a 5V source and a potentiometer only for milestone 1. Starting with the sensor voltage, we created a voltage divider using a potentiometer in order to simulate our CO gas sensor. By adjusting the potentiometer, we can change the two resistor values, causing an output voltage from anywhere between 0 and 5V. We explained this relationship in the building block design equation 3. After this, we designed an RC low pass filter in order to get rid of possible noise in the sensor voltage output. This filter is low pass thanks to the property of a capacitor which prevents voltage from changing instantaneously like from high frequency signals and can only show change from low frequency signals.

Alongside this input block, we used a potentiometer to set the reference voltage. Since we couldn't use a CO sensor at this stage due to parts not arriving, using a potentiometer would allow us to change the reference voltage to whatever the real reference voltage value will be when 50ppm is detected by the sensor. 50ppm is based on our research on OSHA standards for safe levels of CO in a workplace. In future milestones, we will test the output voltage of the sensor when detecting 50 ppm and set that as our reference voltage. However, for this milestone, we simply chose a reference voltage to be 2.5V while we waited for the part. After deciding on this value, we were able to calculate what the value of the 2 resistors in the voltage divider needed to be. Based on our calculations, R_1 had to equal R_2 and since we had a 10k ohms potentiometer, each resistor would be set to 5k ohms.

The next building block was the op amp comparator. This was designed to connect our two previous stages together. At this point, we now have a filtered sensor voltage and a reference voltage. Using an op amp, we used it as a comparator to give an different output based on the values of our two voltage inputs. Because we want the light to turn on when the filtered voltage is greater than the reference, we fed the reference voltage to the positive input of the op amp. We then fed the filtered sensor voltage to the negative input of the op amp. Now, using building block design equations 4 and 5 from above, we know when our op amp will output +5V when the filtered sensor voltage is greater than the reference voltage and when the filtered sensor voltage is less than the reference voltage it will output -5V.

To show an output stage, we connected an LED with a current limiting resistor to the comparator output. The LED we used has a forward voltage range from 1.7-2.4V. As stated above, the op amp outputs 5V when the filtered voltage is greater than the reference. This 5V is enough to power our LED. So, to bring it all together, our system is designed so that when the CO sensor (or simulated sensor in milestone 1) detects more than 50 ppm (momentarily 2.5V), it will turn on the LED, indicating there are dangerous levels of CO present.

The overall input and output for milestone one can be found on figures 6 and 7 above. A V_{out} of -5V means that the LED was not on and a V_{out} of 5V means the LED was on.

Our milestone 2 design starts off with the integrator building block. The input for this building block, and thus the entire milestone 2 design, is the output from the op amp comparator from milestone 1. Knowing this, our integrator building block will either be receiving a value of +5V or -5V. Referring to figure 2, this input is connected to a discharge capacitor system as well as an integrator system. If V_{comp} is -5V, meaning there is not dangerous levels of CO, the op amp for the discharge capacitor system will output a positive 5 volts, according to design equation 8. This 5V is connected to NMOS gate. When an NMOS transistor receives a positive voltage, it allows current to flow between the source and the drain.

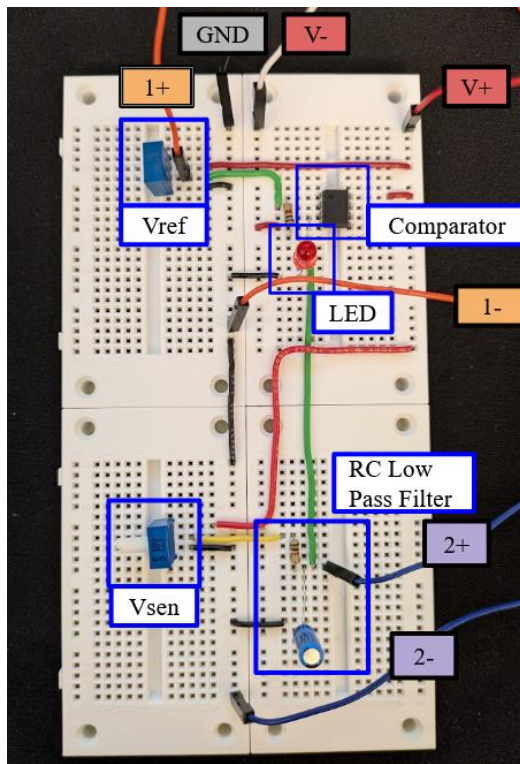
So, since the gate is receiving 5V, current can flow through the transistor which connects the capacitor to ground. This is how we discharge the capacitor. On the other hand, if V_{comp} is +5V, the output of the comparator would be -5V according to design equation 9. This means current will not be able to flow between the source and drain of the transistor. We decided to implement this method of discharging instead of using a large resistor in parallel because it was a much faster method. We wanted our ramping voltage to start at 0 every time no matter what, so the quicker we can discharge the capacitor the better.

Now we will look at the integrating part of this building block. First, the input is brought through a non-amplifying inverter. The reason for this is because our input for the op amp with capacitive feedback is connected to the inverting input. This means if the input into the integrator was 5V, the voltage would steadily decrease from 0 to -5V. This would not give us a PWM output to be used for the fan. So, after implementing this non-amplifying inverter in series with the integrator, we are now able to use design equation 10 to represent the output of the integrator as a function of time. Let's say that there are unsafe levels of CO. The comparator from milestone 1 would output 5V. Using design equation 10 we can calculate the output of the integrator at any time. The output is a linear equation with slope $\frac{5}{10}$ and a y-intercept of 0. This means multiplying the time we want by .5 will give us the output voltage of the integrator at the given time, except the voltage cannot exceed 5V so after it reaches that value, it plateaus until safe CO levels are reached, where it would then very quickly drop back to 0.

The next building block was the astable multivibrator. The function of this building block is to output a PWM like signal to our DC fan. The specific astable multivibrator used in this case was the 555 timer, known for having 3 5k-ohm resistors connected internally that produce a voltage divider network between +Vcc and GND. The resulting output of the non-amplifying inverter is used as the timer's control voltage (CV), which controls the width of the resulting pulse signal and, as such, how much power should go into the resulting fan. Design equations 11 and 12 determine the duty cycle of the circuit, with equation 11 calculating how long the pulse signal remains in a HIGH state and Equation 12 doing the same for a LOW state. Since the output of the non-amplifying inverter is a ramping voltage when CO levels are unsafe, the 555 timer will output a pulse signal with a duty cycle that increases over time until the CV reaches saturation. The pulse signal duty cycle reaches 50% when the control voltage is approximately 2.2651V, with a 66.7% duty cycle being achieved at around 3.333V. The resistors and values were determined so as to have a period most comparable to the 10 seconds it takes for the output of the integrator to reach saturation, as values too large or too small for one of them makes the frequency too fast or too slow to compare.

The final building block was the output block, which contains our DC motor, a flyback diode, and a second NMOS transistor. The output of the astable multivibrator is connected to the NMOS gate connected to ground. As such, any positive voltage that the transistor receives will allow current to flow between the source and the drain and allow the circuit to conduct. This makes sure the fan only turns on when the pulse signal is in a HIGH state, so as to not waste energy. Since the fan is an inductor, however, a lingering backwards current can form from repeated flyback caused by the pulse signal. To eliminate this current, a diode is placed with reverse polarity to the power supply – the pulse signal in this case - and parallel to the inductor. When there's no power coming from the astable multivibrator, the diode reroutes forward any lingering current captured by the inductor through it thanks to its minimal resistance.

Here is some information on our experimental data from MS1:



Each part of our circuit is labeled with blue boxes to indicate what components are in each part.

A guide to the labels:

$V_- = -5V$,

$V_+ = +5V$,

1+/- is measurement channel 1 on the oscilloscope,

2+/- is measurement channel 2 on the oscilloscope,

GND is ground

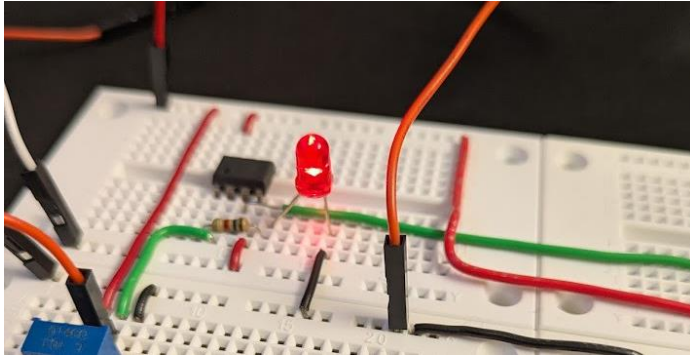
Channel 1 is measuring the reference voltage

Channel 2 is measuring the filtered sensor voltage.

When we turn the knob to have $V_{sensor} > V_{ref}$ representing the case where there is more than 50ppm as the CO measurement meaning it is unsafe to work in the condition:

Period: --	Period: --
Frequency: --	Frequency: --
Peak-peak: 76.256 mV	Peak-peak: 61.005 mV
Mean: 2.490 V	Mean: 3.714 V

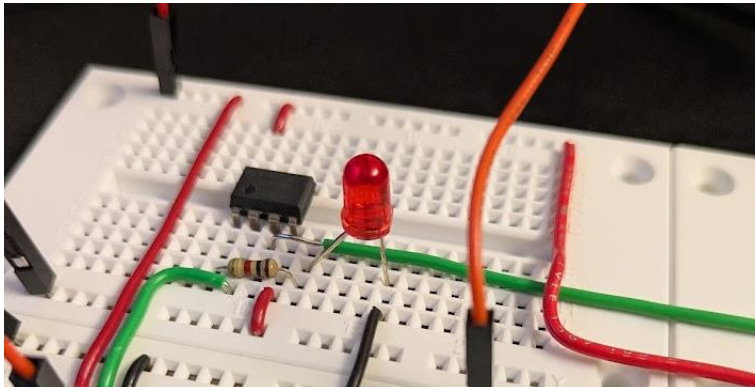
The LED turns on:



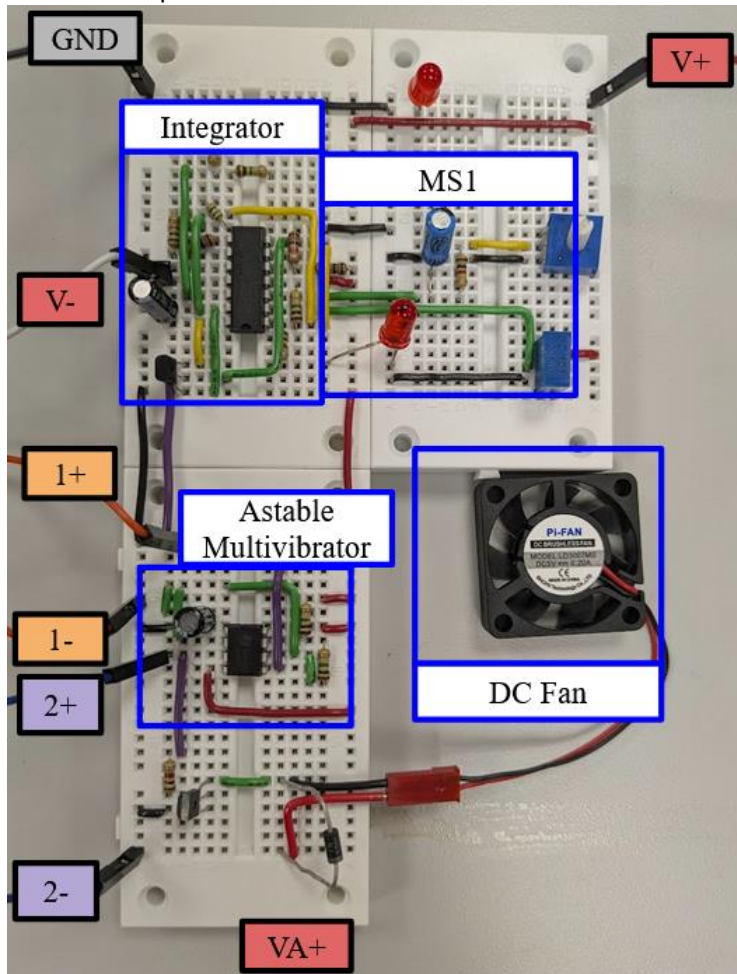
And when we turn the sensor knob the other way to have $V_{sensor} < V_{ref}$ representing the case where there is less than 50ppm as the CO measurement of the room meaning it is safe to work in:

Period: --	Period: --
Frequency: --	Frequency: --
Peak-peak: 76.256 mV	Peak-peak: 76.256 mV
Mean: 2.498 V	Mean: 1.117 V

The LED turns off:



And some experimental data from MS2:



At the top right is MS1 which is described before this page,

The other boxes represent the blocks of MS2.

V+ = +5V

V- = -5V

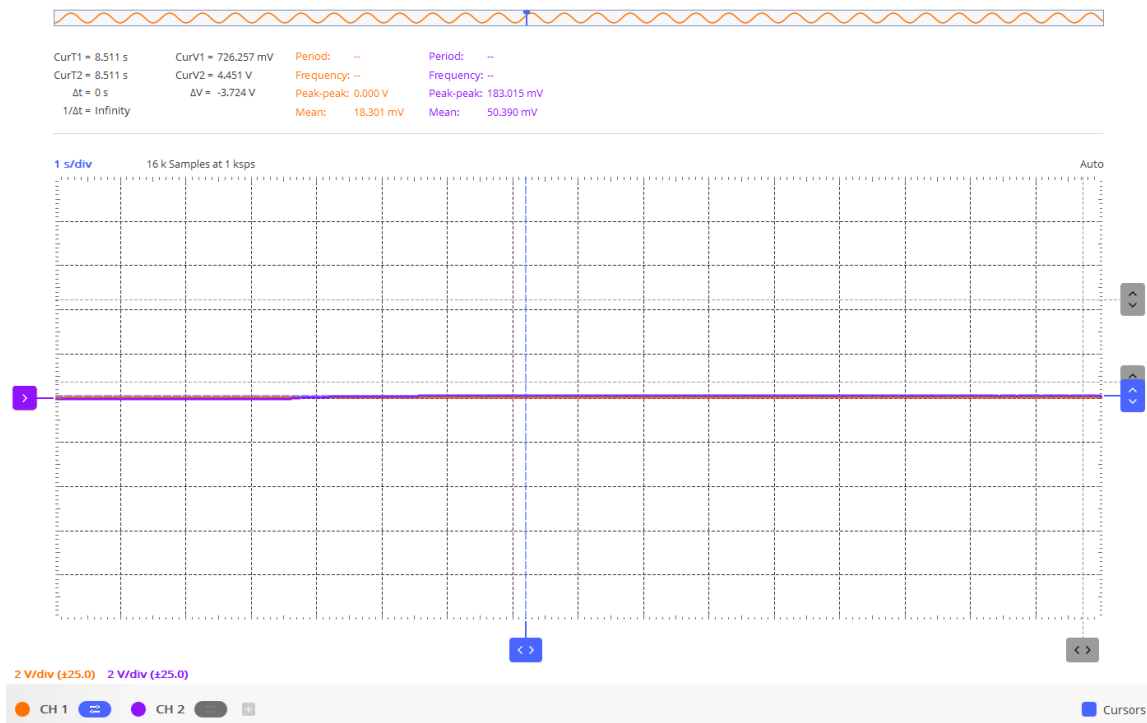
VA+ = +5V from Arduino (for current to power DC fan)

GND is ground

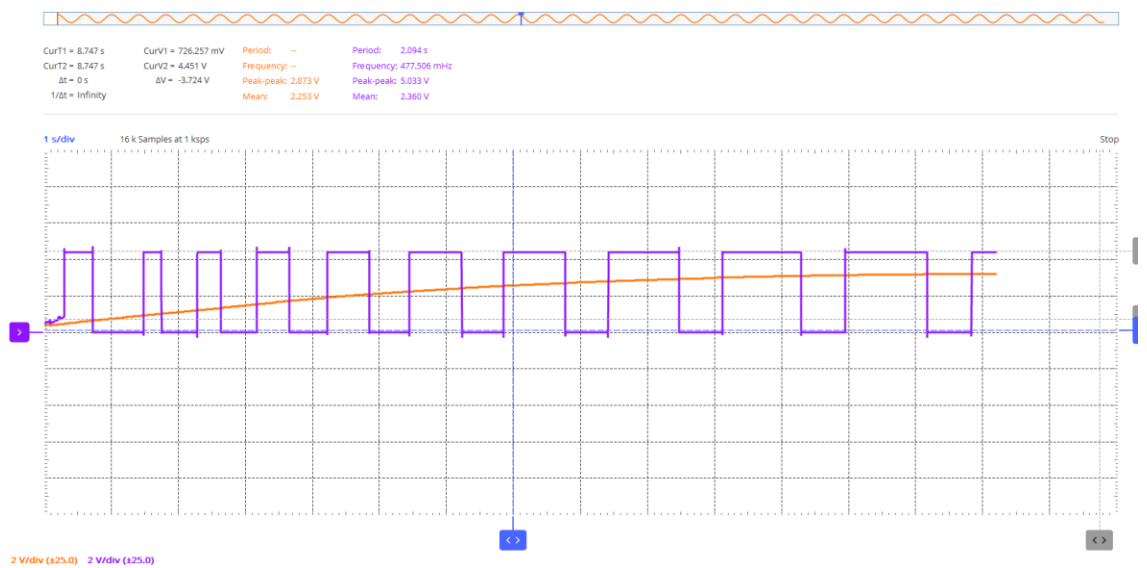
1+/- measures the ramping voltage output of the integrator

2+/- measures the pulse output of the astable multivibrator

When having $V_{sensor} < V_{ref}$ so that V_{comp} is LOW, both V_{int} and V_{pulse} stay at 0V:



Then when $V_{sensor} > V_{ref}$ so that V_{comp} is HIGH, then V_{int} is ramping up and V_{pulse} is increasing in duty cycle:



Operating Conditions

In MS1, one limitation in our circuit comes from the output voltage of our comparator. Since the op-amp we are using isn't rail-to-rail, it won't reach the saturation values. While it wasn't a big problem in MS1 since the output just relied on either a positive or negative voltage, now that we are using specific voltage values in MS2, we had to switch out the op amp from the TL072IP to OP484 which is rail-to-rail.

Another limitation is the temperature/humidity characteristics of our MQ-7 CO sensor. Based on the data sheet, different environmental conditions can cause the properties of our sensor to shift. The manufacturer did testing from -20 to 50 degrees Celsius. So, this means that in different conditions, our reference sensor voltage will have to be different to achieve the 50 ppm that we want. This might become a problem as it continues to get colder and the only way we have of getting CO right now is from our car. We will try to think of another method to get CO to try and prevent this issue.

One tradeoff is the sensitivity of our circuit for some stability. When we implement our CO sensor, we will set the reference voltage for 50 ppm. If the CO sensor is hovering around that range, even small noise could cause the voltage to continuously cross and recross the reference value. This would show up as a noisy comparator output as well as a flickering LED. To prevent this, we could use a schmitt trigger but the tradeoff with this is that if the gap is too wide, the led (or ventilation system in future milestones) would stay on longer than needed. This would waste energy and our whole goal with milestone 2 is to conserve energy by the ventilation only being on when necessary.

One thing we could do to improve our circuit is to possibly use a different capacitor value in our RC circuit. We could weigh our priorities and choose a capacitor value based on that. If we want quicker detection time and are okay with increasing the risk of flickering caused by noise, we could use a lower capacitor value. If we want to prioritize preventing flickering with the cost of slightly delaying the response, we could use a higher capacitor value.

In MS2, one limitation in our circuit comes from tolerances of the RC components. Since we are using such a large resistor in our integrator, the tolerance can drastically affect the value. This can really affect the slope of our ramp. Luckily, right now we don't need the slope to be an exact value, but it is something to consider.

Another limitation has to do with noise from the comparator in MS1. If the output of the comparator contains even just small amounts of noise, the integrator will integrate that error over time, causing the error to be much more exaggerated than it actually is. This will end up affecting the output of the integrator, and in turn, the rest of milestone 2.

One tradeoff is we decided to use a ramping voltage from an integrator instead of using a PID control system in order to save resources and lower costs. The reason we chose this tradeoff is because the levels of CO don't fluctuate rapidly, it's more of a slow steady change. Because of this, we figured that a PID control wouldn't be as important as it could be in other situations. By choosing this tradeoff, we sacrifice some proportional control, as well as a little bit of predictions, but in this case where levels change slowly, it's a tradeoff we were willing to make.

One thing we noticed during testing and during the presentation is that at lower duty cycles, instead of just spinning slower, the fan would just spin and then stop and repeat. One way we could improve our circuit is by changing this. One thing we could do is set a minimum duty cycle for the fan. For example, if we notice that this problem stops happening once we reach 30% duty cycle, we could implement a way so that the fan only starts spinning at 30% duty cycle.

Engineering Design Considerations

One of our design considerations was around our reference voltage being used to compare with the CO sensor. We did some research on the threshold for CO levels in workplaces and found information from OSHA that said 50ppm was the threshold value for when CO levels become unsafe in the workplace and violate health and safety regulations. Another design consideration we actually based our milestone 2 objective on was saving on electricity and money used for HVAC ventilation systems. To prevent wasted electricity and therefore higher costs of operation, we created a system that controls both the on/off condition of the fan and the fan speed itself based on how long the sensor is on.

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Appendix